

UCS321: AI For Engineers

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Course Objective: Develop foundational AI knowledge while equipping participants with practical problem-solving skills using data relevant to engineering applications.

Syllabus:**Modules & Content****1. Introduction to AI for Engineers**

- Definitions: Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Reinforcement Learning (RL)
- Understanding AI's role in engineering decision-making
- Illustrative examples from engineering fields (e.g., predictive maintenance, process control)

2. Data and Preprocessing

- Types of data: sensor data, image data, text data, and time-series data
- Techniques for cleaning, normalization, and handling missing values
- Introduction to feature selection methods

3. Supervised Learning

- Core algorithms: Linear Regression, Logistic Regression, Decision Trees, Random Forest
- Applications: Predicting equipment failure or component lifespan

4. Unsupervised Learning

- Clustering techniques: K-Means, DBSCAN
- Dimensionality reduction methods: Principal Component Analysis (PCA), t-SNE
- Applications: Process classification and anomaly detection

5. Neural Networks & Deep Learning (Intro)

- Fundamentals of Neural Networks: Multilayer Perceptron (MLP), Convolutional Neural Networks (CNN)
- Use cases: Image recognition (e.g., detecting cracks in structures)

6. Time-Series Analysis

- Forecasting techniques: ARIMA and introductory Long Short-Term Memory (LSTM)

models

- Applications: Load prediction and vibration analysis

7. Toolchain for Engineers

- Programming tools: Python, Jupyter Notebook
- Libraries: pandas, scikit-learn, TensorFlow/Keras
- Version control: Using Git effectively

8. Mini Project

- Choose an engineering-specific dataset
Apply preprocessing, modeling, and evaluation techniques

Laboratory Work:

- Build a regression model to predict temperature rise in a motor
- Use K-Means clustering to group materials based on properties
Create a neural network to detect faulty circuits

Case Studies:

- Siemens: AI applications in industrial automation
- General Electric (GE): AI-driven turbine health monitoring
Honeywell: Predictive maintenance in smart factories

Course Learning Objectives (CLO)

The students will be able to:

1.

Text Books

1. Mitchell M., T., Machine Learning, McGraw Hill (1997) 1st Edition.
2. Alpaydin E., Introduction to Machine Learning, MIT Press (2014) 3rd Edition.
3. Vijayvargia Abhishek, Machine Learning with Python, BPB Publication (2018)

Reference Books

1. Bishop M., C., Pattern Recognition and Machine Learning, Springer-Verlag (2011) 2nd Edition.
2. Michie D., Spiegelhalter J. D., Taylor C. C., Campbell, J., Machine Learning, Neural and Statistical Classification. Overseas Press (1994).
3. Bishop, C. M., & Bishop, H. (2023). Deep learning: Foundations and concepts. Springer.
4. Géron, A. (2019). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems (2nd ed.). O'Reilly Media.
5. Liu ZL (2025). Artificial Intelligence for Engineer, (1st ed.) Springer Nature Switzerland.